17 D For a standing wave on a string fixed at both ends, the resonance frequencies are given by :

$$f_n = \frac{nv}{2L}$$

For the 50 cm string, if resonance occurs : $f_n = \frac{n220}{2(0.50)} = 440 \Rightarrow n = 2$

The 50 cm string can cause the tuning fork to resonate.

For the 40 cm string: $f_n = \frac{n220}{2(0.40)} = 275n = 440 \Rightarrow n = 1.6$.

Hence, 440 Hz is not one of the harmonics produced by the 40 cm string, hence it fails to resonate the tuning fork.

(Quick check for 40 cm string)

- Fundamental frequency (n=1): 275 Hz ;
- Second harmonic (n = 2): 275 x 2 = 550 Hz.